# Real Estate Price Forecasting

## Machine Learning for Property Valuation & Investment Strategy

Felipe Taha Sant'Ana

March 2026

# Agenda


1. Business Context
2. Dataset Overview
3. Exploratory Data Analysis
4. Statistical Analysis
5. Model Development & Results
6. Investment Strategy
7. Conclusions & Future Work

# Why Automated Valuation Matters

## The Challenge

- Traditional appraisals take **2–4 weeks** and cost $300–500
- Human bias introduces **up to 10% variance** between appraisers
- Market conditions change faster than manual processes

## Target Metrics

| KPI | Target |
|---|---|
| $R^2$ Score | $> 0.90$ |
| MAPE | $< 12\%$ |
| Latency | $< 200$ms |

## Our Solution

A machine learning model that provides **instant, objective valuations** with less than **11% average error**, enabling faster decisions for buyers, sellers, and lenders.

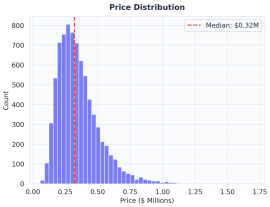
# Dataset at a Glance

## Key Statistics

- **8,000** real estate transactions
- **20** features (property, location, temporal)
- Sales from **2019 Q1 to 2025 Q4**
- **8 neighborhoods** across metro area
- Price range: $57K – $1.7M
- Median price: $320,800

## Feature Groups

| | |
|---|---|
| Property | SqFt, Bedrooms, Year Built, Type |
| Location | Neighborhood, Metro dist., Crime |



# Price by Neighborhood

## Key Findings

- **Waterfront** commands the highest premium (1.5× base)
- **Downtown & Tech Hub** follow as high-value areas
- **Suburbs & Old Town** are the most affordable
- ANOVA: $p < 0.001$ confirms significant price differences

# Price vs. Square Footage

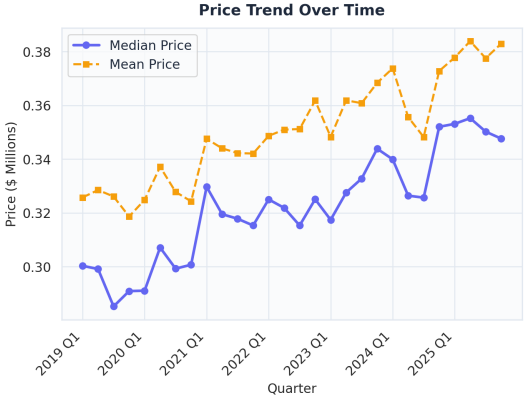
## Analysis

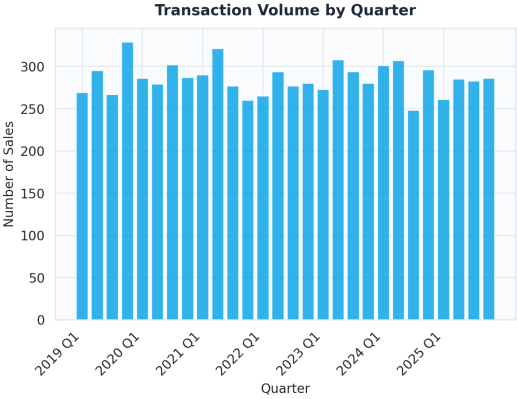
- Strong positive correlation ($r = 0.78$)
- Non-linear relationship — polynomial fit captures the curve
- Color gradient shows bedroom count adds incremental value

## Insight

Each additional 100 sqft adds approximately $15K–25K in value, with diminishing returns above 4,000 sqft.

# Price Trends & Transaction Volume



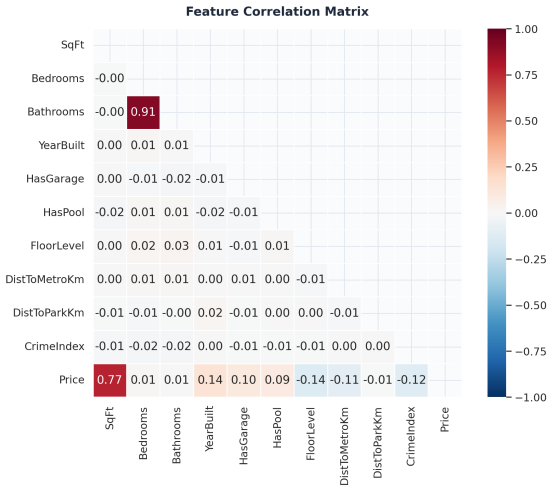

## Price Appreciation

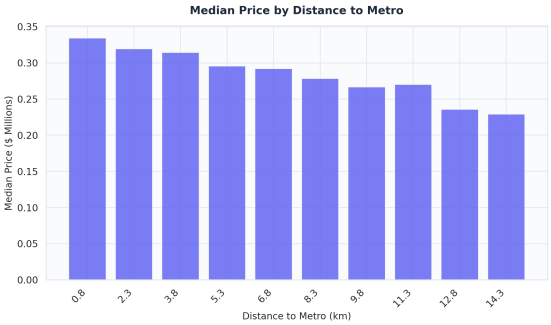
Steady upward trend with **3–4% annual appreciation** across the metro area.

## Volume

Relatively stable transaction volume indicates a healthy, liquid market.

# Correlation Matrix & Metro Proximity




## Metro Premium

# Hypothesis Testing Summary

| Test | Statistic | p-value | Effect |
|---|---|---|---|
| ANOVA: Price $\sim$ Neighborhood | F-test | $p < 0.001$ | Large ($\eta^2$) |
| Pearson: SqFt vs. Price | $r = 0.78$ | $p < 0.001$ | Strong |
| Spearman: Metro Dist. vs. Price | $\rho = -0.15$ | $p < 0.001$ | Moderate |
| Welch's t: Garage premium | t-test | $p < 0.001$ | Significant |
| Kruskal-Wallis: Property Type | H-test | $p < 0.001$ | Significant |

## Conclusion

All hypothesized price drivers show **statistically significant** relationships. Square footage and neighborhood are the dominant factors, followed by proximity to transit and property amenities.

# Modeling Approach

**8,000 Transactions** → **Feature Engineering** → **Log-Price Transform** → **5 Models Trained** → **5-Fold Evalua**

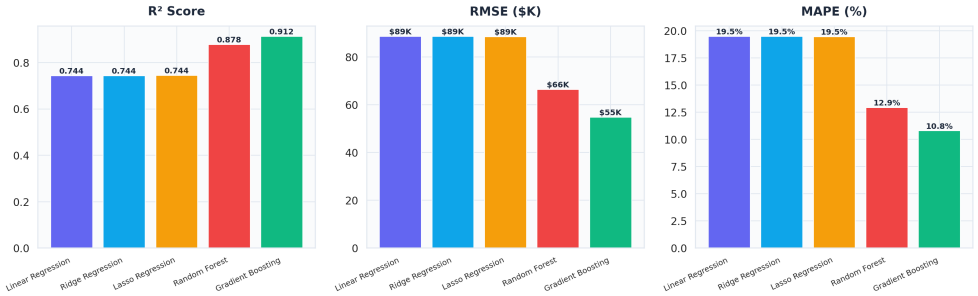
## Engineered Features

- `Age` — property age in years
- `RoomDensity` — rooms per sqft
- `TotalAmenities` — amenity count
- `MetroAccess` — inverse metro dist.

## Models Compared

1. Linear Regression
2. Ridge Regression ($\alpha = 1.0$)
3. Lasso Regression ($\alpha = 0.001$)
4. Random Forest (200 trees)
5. **Gradient Boosting** (200 trees)

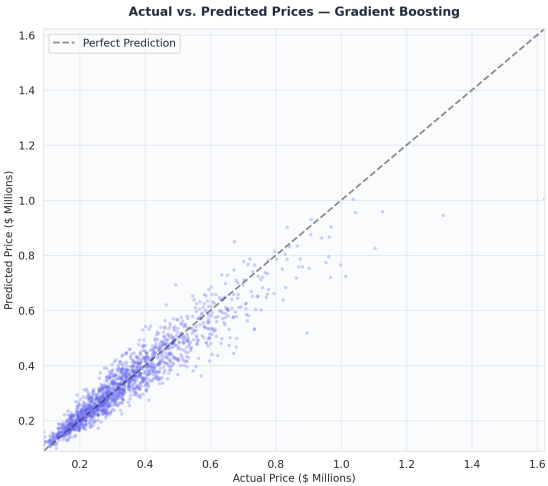
# Model Comparison

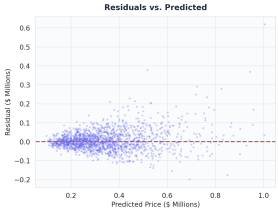


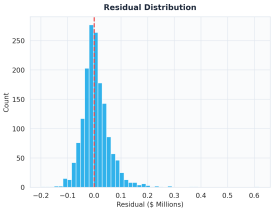

**Winner: Gradient Boosting**

$R^2 = 0.913$ RMSE = \$54,724 MAE = \$35,281 MAPE = 10.8%

# Actual vs. Predicted & Residuals

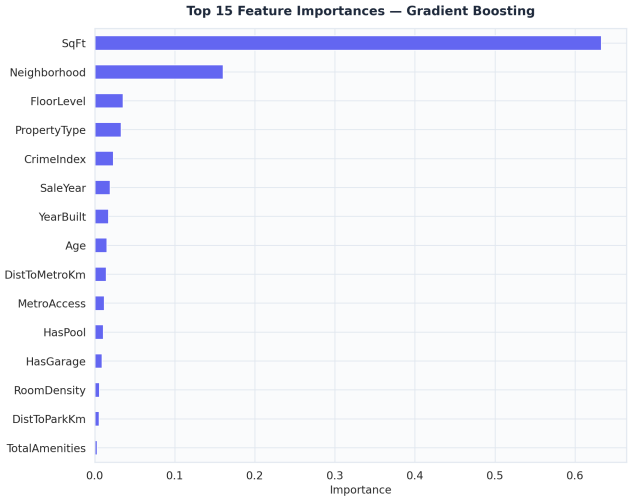



## Diagnostic Summary

# Feature Importance



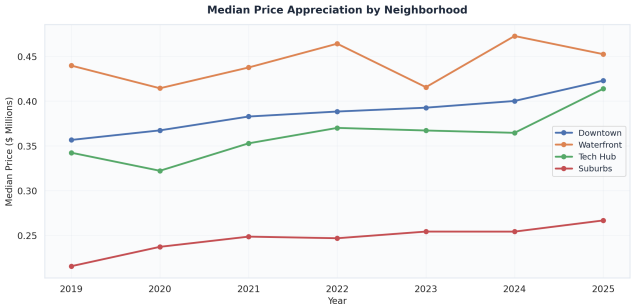

## Top Price Drivers

1. **Square Footage** — dominant predictor
2. **Neighborhood** — location premium
3. **Metro Access** — transit proximity
4. **Year Built** — recency premium
5. **Total Amenities** — garage, pool, elevator

## Takeaway

Physical size and location account for the majority of price variance —

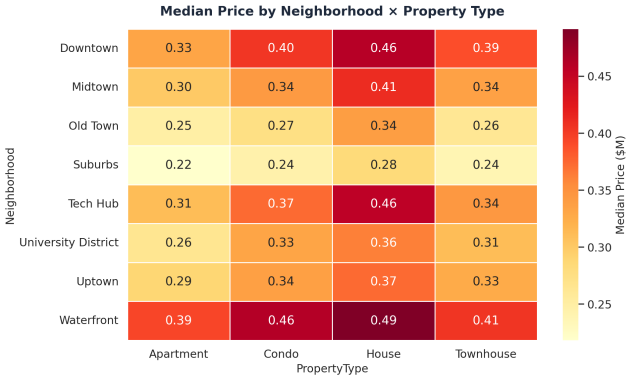
# Neighborhood Appreciation Trends




## Growth Leaders

| | |
|---|---|
| Waterfront | Highest absolute appreciation, limited supply |
| Tech Hub | Fastest relative growth, driven by employment |
| Downtown | Steady premium, mature market |
| Suburbs | Affordable entry, remote-work tailwind |

# Investment Matrix




## Strategy Recommendations

**Growth** Tech Hub condos & apartments — ride the appreciation wave

**Value** Suburb houses — best sqft/dollar with upside

**Premium** Waterfront houses — trophy assets with stability

**Yield** University District apartments — high rental demand

## Risk Factors

# Summary of Results

## Achievements

- Gradient Boosting: $\mathbf{R^2 = 0.913}$
- Mean error: **$35K** (10.8% MAPE)
- All targets met ($R^2 > 0.90$, MAPE $< 12\%$)
- Identified top 5 price drivers
- Produced actionable investment insights

## Future Roadmap

1. **Q2 '26:** SHAP explainability dashboard
2. **Q3 '26:** Geospatial kriging models
3. **Q3 '26:** Image-based CNN valuation
4. **Q4 '26:** FastAPI deployment
5. **Q1 '27:** Time series forecasting with LSTM

# Thank You

Questions & Discussion

**GitHub:** github.com/ftahas
**Email:** ftahas@proton.me
**LinkedIn:** linkedin.com/in/ftsantana